

Notes: Hutton, Marcus, and Tehranian (JFE, 2009)
Opaque financial reports, R^2 , and crash risk

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1 Big picture

- **Research question.** Does financial-reporting opacity affect the distribution of firm-level stock returns? More specifically, do opaque financial reports make stock prices more synchronous with the market and make firms more prone to sharp negative crashes?
- **Main idea.** Managers can use earnings management to delay the release of bad firm-specific information. When the accumulated bad news can no longer be hidden, the information is released at once and the stock price crashes.
- **Theory being tested.** The paper is motivated by the Jin and Myers information-hoarding model. In that model, opacity lowers the flow of firm-specific information into prices during normal times, but increases the probability of sudden crashes when hidden bad news becomes public.
- **Empirical proxy for opacity.** The paper measures opacity with *OPAQUE*, the prior three-year moving sum of the absolute value of discretionary accruals. Large absolute discretionary accruals are interpreted as evidence of earnings management and therefore less transparent reporting.
- **Main finding on synchronicity.** More opaque firms have higher R^2 from an expanded market-and-industry return model. Equivalently, they have lower firm-specific return variation relative to market and industry variation.
- **Main finding on crashes.** More opaque firms are more likely to experience firm-specific crashes. The effect is statistically strong and economically meaningful.
- **Asymmetry.** Opacity predicts negative crashes, but it does not predict large positive jumps. This asymmetry is central: the paper is not simply saying that opaque firms have fatter tails; it is saying that opacity is connected specifically to bad-news stockpiling.
- **Post-SOX result.** The relation between discretionary-accrual opacity and both R^2 and crash risk largely disappears after the Sarbanes-Oxley Act. This is consistent with SOX reducing earnings-management opacity, or reducing firms' ability to hide information.
- **Economic takeaway.** Financial-reporting opacity affects not only accounting quality but also the shape of stock return distributions. Poor transparency can make prices look calm most of the time while increasing left-tail risk.

2 Incremental contribution of the paper

- **From country-level opacity to firm-level opacity.** Prior work, especially Jin and Myers, studies opacity across countries. Hutton, Marcus, and Tehranian construct a firm-year opacity measure and test whether the same logic holds within the U.S. market.
- **Direct link between accounting and crash risk.** The paper connects an accounting-based measure of earnings management to a finance outcome: the probability of a stock-price crash.
- **Extends the R^2 literature.** Existing work often interprets low R^2 as greater firm-specific information in prices. This paper shows that a firm-level opacity proxy is associated with higher R^2 , supporting the transparency interpretation.

- **Tests left-tail risk rather than only volatility.** The contribution is not only about the second moment of returns. The paper studies extreme negative residual returns and contrasts them with symmetric positive jumps.
- **Validation of the opacity proxy.** The paper shows that firms with higher *OPAQUE* are more likely to restate financial statements, supporting the interpretation that discretionary accruals capture less-than-transparent reporting.
- **SOX as a useful timing test.** The post-SOX analysis asks whether the opacity proxy loses predictive power after a major regulatory shock that raised the cost of earnings manipulation.
- **Risk-management implication.** The results imply that accounting opacity can help identify firms with greater latent crash risk, which matters for portfolio risk, tail-risk management, and option pricing.

3 Data and measurement

3.1 Sample construction

The paper combines weekly stock returns from CRSP with annual accounting data from Compustat.

- **Period.** Fiscal years 1991–2005.
- **Final sample.** 40,882 firm-years.
- **Major exclusions.** The paper excludes firm-years with incomplete stock-return data, financial firms, utilities, firm-years with insufficient control-variable data, and firm-years without enough data to calculate three lags of discretionary accruals.
- **Industry scope.** The final sample covers 43 of the 49 Fama-French industries.
- **Reason for starting in 1991.** The authors need three annual lags of discretionary accruals computed using the Statement of Cash Flow method.

Interpretation. The dataset is a broad U.S. firm-year panel that allows the authors to connect accounting opacity to return synchronicity and tail events.

3.2 Opacity from discretionary accruals

The key explanatory variable is *OPAQUE*. It is based on the modified Jones model. The first step estimates normal accruals within each Fama-French industry-year:

$$\frac{TA_{jt}}{Assets_{j,t-1}} = a_0 \frac{1}{Assets_{j,t-1}} + b_1 \frac{\Delta Sales_{jt}}{Assets_{j,t-1}} + b_2 \frac{PPE_{jt}}{Assets_{j,t-1}} + \varepsilon_{jt}.$$

Discretionary accruals are then computed using the modified sales term:

$$DiscAcc_{jt} = \frac{TA_{jt}}{Assets_{j,t-1}} - \left(\hat{a}_0 \frac{1}{Assets_{j,t-1}} + \hat{b}_1 \frac{\Delta Sales_{jt} - \Delta Receivables_{jt}}{Assets_{j,t-1}} + \hat{b}_2 \frac{PPE_{jt}}{Assets_{j,t-1}} \right).$$

The opacity measure is:

$$OPAQUE_t = |DiscAcc_{t-1}| + |DiscAcc_{t-2}| + |DiscAcc_{t-3}|.$$

- The absolute value is used because both large positive discretionary accruals and later reversing negative discretionary accruals are informative about earnings management.
- The three-year moving sum captures persistent reporting behavior rather than a one-year shock.
- The measure is firm-specific, unlike country-level opacity variables used in much of the earlier literature.

Interpretation. A high value of *OPAQUE* means the firm has repeatedly large abnormal accruals, suggesting that financial statements reveal less firm-specific information.

3.3 Firm-specific return variation and R^2

The paper estimates an expanded index model using weekly returns:

$$r_{j,t} = \alpha_j + b_{1,j}r_{m,t-1} + b_{2,j}r_{i,t-1} + b_{3,j}r_{m,t} + b_{4,j}r_{i,t} + b_{5,j}r_{m,t+1} + b_{6,j}r_{i,t+1} + \varepsilon_{j,t},$$

where r_m is the CRSP value-weighted market return and r_i is the Fama-French value-weighted industry return.

The measure of firm-specific information in returns is:

$$IDIOSYN = \ln \left(\frac{1 - R^2}{R^2} \right).$$

- A high R^2 means the stock moves more with market and industry returns.
- A high *IDIOSYN* means more idiosyncratic firm-specific return variation.
- If opacity blocks firm-specific information from prices, *OPAQUE* should be negatively related to *IDIOSYN* and positively related to R^2 .

3.4 Crash and jump measures

The residuals from the index model are transformed into firm-specific weekly returns:

$$FirmSpecificReturn_{j,t} = \ln(1 + \hat{\varepsilon}_{j,t}).$$

- **Crash.** *CRASH* equals one if a firm-year has at least one firm-specific weekly return 3.09 standard deviations below that firm's fiscal-year mean.
- **Jump.** *JUMP* equals one if a firm-year has at least one firm-specific weekly return 3.09 standard deviations above that firm's fiscal-year mean.
- The 3.09 cutoff corresponds to a 0.1 percent tail event in a normal distribution.
- Events are defined using residual returns rather than raw returns, so broad market declines do not mechanically generate crashes.

Interpretation. Crashes and jumps are symmetric in construction. Therefore, if opacity predicts only crashes, the result supports the bad-news-hoarding mechanism rather than a generic fat-tail explanation.

3.5 Controls and validation

The main control variables include size, market-to-book, leverage, return on equity, industry return variance, skewness, and kurtosis.

The paper also validates *OPAQUE* using financial restatements. Firms with higher opacity are more likely to later restate financial statements, and restatement firm-years have substantially higher average opacity.

Interpretation. The restatement evidence supports the view that *OPAQUE* captures opacity and not merely innocuous accounting variation.

4 Methodology and empirical specifications

4.1 R^2 and idiosyncratic information regressions

The first main empirical test is an OLS regression of *IDIOSYN* on opacity:

$$IDIOSYN_{j,t} = \alpha + \beta_1 OPAQUE_{j,t} + \beta_2 OPAQUE_{j,t}^2 + \gamma' X_{j,t} + u_{j,t}.$$

- The dependent variable is decreasing in R^2 .
- A negative coefficient on *OPAQUE* means opaque firms have higher R^2 and less firm-specific information in returns.
- The squared term allows the relation to be nonlinear.
- The paper uses robustness checks that alter the return window, the accrual measure, the return model, and the estimation method.

Interpretation. The main coefficient is not a structural estimate of hidden information. It is a reduced-form relation showing that earnings-management opacity is associated with lower firm-specific return variation.

4.2 Crash-risk logit regressions

The second main empirical test uses logit regressions for crash probability:

$$Pr(CRASH_{j,t} = 1) = F\left(\alpha + \beta_1 OPAQUE_{j,t} + \beta_2 OPAQUE_{j,t}^2 + \gamma' X_{j,t}\right).$$

- A positive *OPAQUE* coefficient means opaque firms are more crash-prone.
- The squared term captures diminishing marginal effects at high levels of opacity.
- The paper reports marginal impacts from moving variables from the 25th to the 75th percentile.
- The baseline model finds that moving opacity from the 25th to the 75th percentile increases crash probability by about 1.56 percentage points.

Interpretation. This is the key tail-risk result: opaque reporting predicts the realization of large negative firm-specific return shocks.

4.3 Positive-jump placebo-style test

The paper repeats the crash analysis using *JUMP* as the dependent variable:

$$Pr(JUMP_{j,t} = 1) = F\left(\alpha + \beta_1 OPAQUE_{j,t} + \beta_2 OPAQUE_{j,t}^2 + \gamma' X_{j,t}\right).$$

- If opacity merely proxyied for fat-tailed return distributions, it should also predict positive jumps.
- It does not.
- The coefficients on *OPAQUE* and *OPAQUE*² are economically small and statistically insignificant in the jump regressions.

Interpretation. The absence of a positive-jump result supports the mechanism that managers hide bad news, not good news.

4.4 SOX interaction tests

The paper then tests whether the opacity relations weaken after Sarbanes-Oxley:

$$Outcome_{j,t} = \alpha + \beta_1 OPAQUE_{j,t} + \beta_2 SOX_t \times OPAQUE_{j,t} + \gamma' X_{j,t} + u_{j,t},$$

with analogous terms for *OPAQUE*².

- *SOX* equals one in 2002 and afterward.
- In the *IDIOSYN* regressions, the post-SOX interaction has the opposite sign from the pre-SOX opacity coefficient.
- In the crash regressions, the post-SOX interaction terms also offset the pre-SOX opacity relation.
- Opacity does not predict positive jumps in either the pre-SOX or post-SOX periods.

Interpretation. The SOX evidence is consistent with the view that discretionary accruals became a weaker proxy for hidden information after stronger accounting oversight.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies robust conditional associations between an accounting-based opacity proxy and stock-return outcomes. The evidence is strongest as firm-year panel evidence consistent with an information-hoarding mechanism.

5.2 Main hypotheses

- **H1.** Opaque financial reports are associated with more market-synchronous returns.
- **H2.** Opaque financial reports are associated with greater crash risk.

- **H3.** Opaque financial reports are not associated with greater positive-jump risk.

The empirical results support all three hypotheses.

5.3 Why discretionary accruals matter

Discretionary accruals are useful because managers have discretion over accrual estimates and can use that discretion to manage reported earnings. Large abnormal accruals are therefore a plausible signal that the firm is managing information flow to investors.

5.4 Why crashes are defined from residuals

Using raw returns would mechanically create many crashes in market downturns. The paper instead uses residuals from a market-and-industry model to isolate firm-specific extreme events.

5.5 Key endogeneity concern

The paper is not a randomized experiment. Firms that manage earnings may differ from other firms in ways that also affect stock returns. The authors address this concern through controls, size-by-opacity sorts, restatement validation, robustness checks, and the SOX timing test. These checks strengthen the interpretation but do not turn the design into a clean experiment.

5.6 Economic interpretation of the asymmetry

The contrast between crashes and jumps is central. Managers have incentives to hide bad news to protect compensation, financing opportunities, reputation, or job security. They have weaker incentives to hide good news. Therefore, opaque firms should experience sudden negative releases, not symmetric positive releases.

6 Tables and figures

6.1 Table 1 and Figure 1: Sample construction and annual accrual behavior

Read:

- Table 1 shows the construction of the 40,882 firm-year sample.
- The starting Compustat universe has 109,026 firm-years from 1991 to 2005.
- The largest exclusions are financial services and utilities, incomplete stock-return data, and insufficient data for accrual calculations.
- Figure 1 plots the mean absolute value of annual discretionary accruals and market returns by year.
- Annual absolute discretionary accruals are usually between 6 percent and 8 percent of lagged assets, but are higher in several years around the late 1990s and early 2000s.

Economic message: The sample is broad enough for firm-year panel tests, and discretionary accruals vary meaningfully over time.

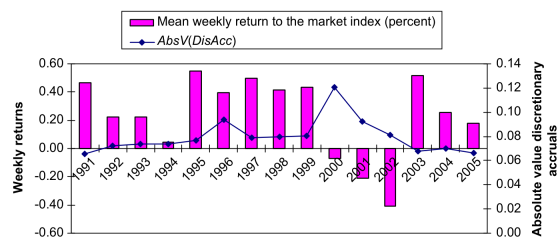
Table 1

Sample development, industry membership, and fiscal years of sample.
Source of data in Panel B is http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/det_49_ind_port.html.

Panel A: Sample development		Number of firm years	
All Compustat firm fiscal years 1991–2005		109,026	
Excluding firm fiscal years:			
With incomplete stock return data		22,777	
Financial services and utilities		28,051	
With insufficient financial data to calculate control variables		7,413	
With insufficient financial data to calculate three lags of discretionary accruals		9,903	
Final sample		40,882	
Panel B: Fama and French industries			
Industry	Number of firm years	Industry	Number of firm years
Aircraft	242	Consumer Goods	922
Agriculture	181	Measuring & Control Equipment	1,188
Automobiles & Trucks	882	Machinery	1,870
Beer & Liquor	181	Restaurants, Hotels, Motels	910
Construction Materials	1,066	Medical Equipment	1,610
Printing & Publishing	517	Non-Metallic & Industrial Metal Mining	223
Shipping Containers	173	Petroleum & Natural Gas	1,081
Business Services	2,260	Business Supplies	867
Chemicals	1,003	Personal Services	482
Electronic Equipment	3,019	Retail	2,682
Apparel	792	Rubber & Plastic Products	509
Construction	514	Shipbuilding, Railroad Equipment	107
Coal	37	Tobacco Products	53
Pharmaceutical Products	2,370	Candy & Soda	110
Electrical Equipment	846	Computer Software	2,851
Fabricated Products	247	Steel Works, etc.	912
Food Products	936	Communication	1,246
Entertainment	681	Recreation	433
Precious Metals	253	Transportation	1,295
Defense	80	Textiles	320
Computers	1,300	Wholesale	1,889
Healthcare	752		
Total			40,882
Panel C: Observations in each fiscal year		Number of observations	
Fiscal year			
1991		2,022	
1992		2,295	
1993		2,441	
1994		2,557	
1995		2,680	
1996		2,872	
1997		3,023	
1998		2,976	
1999		2,858	
2000		3,053	
2001		2,754	
2002		2,726	
2003		2,842	
2004		3,105	
2005		2,678	
Total		40,882	

ΔSales_{jt} denotes change in sales for firm j in year t , and DisAccrual_{jt} denotes discretionary annual accruals as a fraction of

Table 1: Sample development, Fama-French industry composition, and fiscal-year observations.



lute value of discretionary accruals and mean weekly return on the Center for Research in Security Prices value-weighted n

Figure 1: Annual absolute discretionary accruals and market returns.

6.2 Figure 2 and Table 2: OPAQUE and crash/jump frequency

Read:

- Figure 2 plots the three-year moving sum of absolute discretionary accruals, the paper's main opacity proxy.
- *OPAQUE* is smoother and less tightly tied to contemporaneous market returns than the single-year accrual measure.
- Table 2 reports that 17.1 percent of firm-years have at least one crash and 22.0 percent have at least one positive jump.
- Crash weeks have a mean raw return of -22.74 percent, while jump weeks have a mean raw return of 33.27 percent.

Economic message: Extreme firm-specific return events are common enough to study systematically, and the paper's crash/jump definitions capture economically large weekly moves.

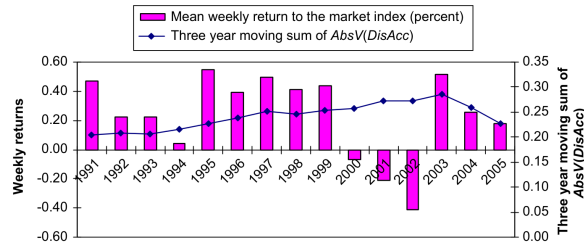


fig. 2. *OPAQUE* and mean weekly return on the Center for Research in Security Prices value-weighted market index.

Figure 2: Three-year opacity measure and market returns.

Table 2
Crashes and jumps, frequency and weekly returns.

Panel A reports the frequency of firm-specific crashes and jumps for 40,882 firm-years in the sample period 1991–2005. Crashes and jumps are defined based on residuals from an expanded index model regression with market and industry returns as explanatory variables. Weekly firm-specific residual returns that are 3.09 standard deviations below or above the mean for the firm's fiscal year are categorized as crashes or jumps.

Panel B reports the mean, median, and variance of raw weekly returns for *CRASH* weeks, *JUMP* weeks, and *ALL OTHER* weeks. The first set of columns presents statistics for individual firms. The middle set of columns presents analogous results for the market index. In this panel, *CRASH* (or *JUMP*) weeks refer to any week in which any firm in the sample crashes (or jumps). The last set of columns report statistics averaging across industries. If any firm in an industry crashes (or jumps) in a given week, that is defined as a crash (or a jump) week for the industry. Weekly returns for 40,882 firm-years in the sample period 1991–2005. See Appendix A for all variable definitions.

Panel A: Crash and jump frequency			Number of observations			Percent of sample				
Frequency										
Crashes in the firm year										
0			33,932					83.0		
1			6,786					16.6		
2			184					0.5		
3			0					0.0		
Total			40,882							
Jumps in the firm year										
0			31,853					77.9		
1			8,767					21.4		
2			261					0.6		
3			1					0.0		
Total			40,882							
Panel B: Return in crash and jump weeks versus all other weeks										
Week classification	Number of observations	Firm returns			Market index			Industry index		
		Mean	Median	Variance	Mean	Median	Variance	Mean	Median	Variance
<i>CRASH</i> Weeks	7,134	-0.2274	-0.2014	0.0147	0.0033	0.0039	0.0004	0.0020	0.0020	0.0010
<i>JUMP</i> Weeks	9,292	0.3327	0.2623	0.1007	0.0055	0.0066	0.0003	0.0084	0.0071	0.0008
<i>ALL OTHER</i> Weeks	2,101,147	0.0026	0.0000	0.0054	0.0023	0.0036	0.0005	0.0025	0.0029	0.0010

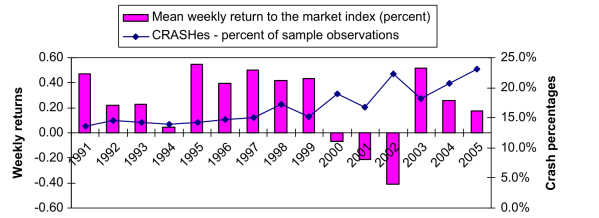
Table 2: Frequency and returns of crashes and jumps.

6.3 Figures 3 and 4: Time-series patterns in crashes and jumps

Read:

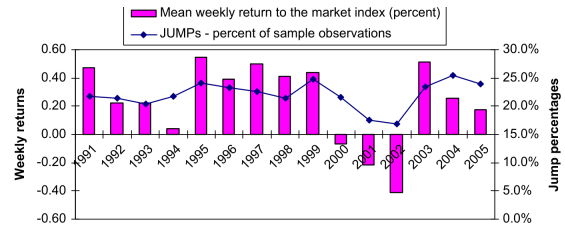
- Figure 3 shows that crash frequency increases in the second half of the sample.
- Figure 4 shows that jump frequency is more stable over time.
- Both figures include market returns, but events are defined from residual returns, so the extreme-event frequency is not mechanically the market's movement.

Economic message: Crash risk and jump risk are not mirror images over the sample period. This motivates separate analysis of negative and positive tail events.



Crash frequency and the mean weekly return on the Center for Research in Security Prices value-weighted market index.

Figure 3: Crash frequency and market returns.



Jump frequency and the mean weekly return on the Center for Research in Security Prices value-weighted market index.

Figure 4: Jump frequency and market returns.

6.4 Table 3: Summary statistics and correlations

Read:

- Panel A reports summary statistics for market value, opacity, R^2 , residual-return volatility, kurtosis, skewness, profitability, valuation, leverage, and industry variance.
- The mean *OPAQUE* value is 0.243, and the mean R^2 is 0.250.
- Panel B shows that many variables are strongly correlated. In particular, size is strongly related to R^2 and to opacity.

- The univariate correlation between opacity and R^2 is not the final test because size and opacity are correlated in opposite ways.

Economic message: Controls are essential. The paper must separate the role of opacity from the role of firm size, volatility, and other return-distribution characteristics.

Table 3
Key variables of interest.
Panel A covers 40,882 firm-years in the sample period 1991–2005. Q1 and Q3 are first and third quartile values. In Panel B, correlations are computed from 40,882 firm-years in the sample period 1991–2005. Pearson correlations are above the diagonal; Spearman correlations below the diagonal; p-values appear below correlations. See Appendix A for variable definitions.

Panel A: Descriptive statistics	Q1	Mean	Median	Q3	Standard deviation
Market value (lagged)	81.2	3,073.4	298.0	1,981.5	14,300.3
Opacity	0.102	0.243	0.174	0.302	0.222
R^2	0.139	0.230	0.209	0.321	0.164
Std Dev [ln(1+residual)]	0.036	0.058	0.052	0.074	0.030
Kurtosis	0.152	1.072	0.878	2.146	2.758
Skewness	-0.258	0.013	0.114	0.525	0.778
ROE	-0.096	-0.011	0.087	0.155	0.428
Market-to-book (lagged)	1.240	3.077	2.628	3.453	3.434
Leverage (lagged)	0.336	0.475	0.464	0.624	0.211
Var(Industry Index)	0.0004	0.0010	0.0007	0.0012	0.0010

Table 3, Panel A: Descriptive statistics.

Panel B: Correlation matrix

Variable	Size	Opacity	R^2	Std Dev [ln(1+resid)]	Kurtosis	Skewness	Crashes	Jumps	ROE	Market to book	Leverage	Variance of Industry Index
Size (lagged)	-0.215 -0.0001	0.580 -0.0001	-0.465 -0.0001	-0.063 -0.0001	-0.210 -0.0001	0.076 -0.0001	-0.171 -0.0001	0.182 -0.0001	0.245 -0.0001	0.142 -0.0001	0.048 -0.0001	
Opacity	-0.282 -0.0001	-0.074 -0.0001	0.358 -0.0001	0.035 -0.0001	0.004 -0.0001	0.014 -0.0001	-0.216 -0.0001	0.182 -0.0001	-0.113 -0.0001	0.128 -0.0001	0.176 -0.0001	
R^2	0.548 -0.0001	-0.100 -0.0001	-0.305 -0.0001	-0.179 -0.0001	-0.117 -0.0001	-0.024 -0.0001	-0.157 -0.0001	0.096 -0.0001	0.151 -0.0001	0.072 -0.0001	0.129 -0.0001	
Std Dev [ln(1+resid)]	-0.511 -0.0001	0.467 -0.0001	-0.311 -0.0001	0.224 -0.0001	0.236 -0.0001	0.082 -0.0001	0.131 -0.0001	-0.368 -0.0001	0.136 -0.0001	-0.129 -0.0001	0.309 -0.0001	
Kurtosis	-0.129 -0.0001	0.070 -0.0001	-0.215 -0.0001	0.224 -0.0001	-0.008 -0.0001	0.430 -0.0001	0.173 -0.0001	-0.064 -0.0001	-0.008 -0.0001	-0.027 -0.0001	0.002 -0.0001	
Skewness	-0.219 -0.0001	0.048 -0.0001	-0.163 -0.0001	0.077 -0.0001	0.086 -0.0001	-0.197 -0.0001	0.156 -0.0001	-0.001 -0.0001	-0.073 -0.0001	0.014 -0.0001	-0.012 -0.0001	
Crashes	0.088 -0.0001	0.010 -0.0001	-0.015 -0.0001	0.082 -0.0001	0.486 -0.0001	-0.365 -0.0001	-0.149 -0.0001	-0.023 -0.0001	0.034 -0.0001	-0.021 -0.0001	0.021 -0.0001	
Jumps	-0.173 -0.0001	0.042 -0.0001	-0.169 -0.0001	0.189 -0.0001	0.135 -0.0001	0.182 -0.0001	-0.140 -0.0001	-0.014 -0.0001	-0.012 -0.0001	-0.003 -0.0001	-0.015 -0.0001	
ROE	0.281 -0.0001	-0.137 -0.0001	0.148 -0.0001	-0.174 -0.0001	-0.300 -0.0001	-0.025 -0.0001	-0.030 -0.0001	-0.148 -0.0001	-0.062 -0.0001	0.034 -0.0001	-0.009 -0.0001	
M/B ratio (lagged)	0.422 -0.0001	0.007 -0.0001	0.242 -0.0001	-0.030 -0.0001	-0.051 -0.0001	-0.131 -0.0001	0.068 -0.0001	-0.197 -0.0001	0.294 -0.0001	0.059 -0.0001	0.090 -0.0001	
Leverage (lagged)	0.136 -0.0001	-0.142 -0.0001	0.061 -0.0001	-0.157 -0.0001	-0.031 -0.0001	0.068 -0.0001	-0.022 -0.0001	-0.004 -0.0001	0.127 -0.0001	-0.020 -0.0001	-0.090 -0.0001	
Variance of Industry Index	0.045 -0.0001	0.170 -0.0001	0.177 -0.0001	0.339 -0.0001	0.072 -0.0001	-0.036 -0.0001	0.022 -0.0001	-0.024 -0.0001	-0.002 -0.0001	0.027 -0.0001	-0.075 -0.0001	

Table 3, Panel B: Correlation matrix.

6.5 Tables 4 and 5: Validation of opacity and size-opacity sorting

Read Table 4:

- Restatements increase monotonically with opacity: 1.20 percent for low-opacity firm-years, 1.63 percent for the middle group, and 1.92 percent for the high-opacity group.
- Restatement firm-years have higher mean *OPAQUE* than non-restatement firm-years: 0.3634 versus 0.2645.
- This supports the interpretation that high *OPAQUE* reflects less transparent reporting.

Read Table 5:

- The sample is sorted into five size quintiles and three opacity groups.
- Small firms are more likely to have high opacity; large firms are more likely to have low opacity.
- Within every size quintile, R^2 is higher for high-opacity firms than for low-opacity firms.

Economic message: The opacity proxy is connected to accounting restatements and to return synchronicity even after conditioning on size.

Table 4
Opacity and financial restatements.

This table shows the relation between opacity and the incidence of restatements of financial statements. Panel A divides firm-years into three groups based on the value of *OPAQUE* and presents the rate of restatements for each group. Panel B compares the levels of *OPAQUE* for firm-years with and without restatements.

Panel A: Incidence of restatements by opacity measure			
Opacity group		Percent of observations with restatements	
Lowest		1.20%	
Middle		1.63%	
Highest		1.92%	
Chi ² statistic for equality across groups		8.35 ($p = 0.0154$)	
Chi ² statistic for equality between high and low opacity groups		8.37 ($p = 0.0038$)	
Panel B: Levels of opacity for restatement and no-restatement firm-years			
	Number of observations	Mean <i>OPAQUE</i>	Median <i>OPAQUE</i>
No restatements	14,437	0.2645	0.1862

Table 4: Opacity and financial restatements.

Table 5
Firm characteristics in groups sorted by size and opacity.

We sort our sample into five size quintiles and into three opacity groups (1 = low; 3 = high). Break points for the two sorts are estimated independently, so the number of observations in each cell differs. Panel A presents the number of observations in each cell. Panel B demonstrates that opacity varies substantially even within size groups. Panel C presents the average R^2 for firm-years in each cell and a test of the hypothesis that R^2 is equal in the high versus low opacity group within each size quintile. There are 40,882 firm-years in the sample period 1991–2005.

Panel A: Observations in each size/opacity group					
Size	Market cap (millions of dollars)	Opacity group			Total
		1	2	3	
1	39	1,568	2,701	3,902	8,171
2	123	2,002	2,772	3,406	8,180
3	331	2,536	2,778	2,865	8,179
4	994	3,249	2,855	2,076	8,180
5	14,318	4,267	2,526	1,379	8,172
Total		13,622	13,632	13,628	
Panel B: Level of opacity for each size/opacity group					
Size	Opacity group				
	1	2	3		
1	0.085	0.184		0.482	
2	0.082	0.181		0.488	
3	0.082	0.178		0.477	
4	0.080	0.176		0.448	
5	0.074	0.175		0.433	
Panel C: R^2 for each size/opacity group					
Size	Opacity group			p -value: (3)–(1)	
	1	2	3		
1	0.147	0.150	0.160	<0.0001	
2	0.177	0.181	0.189	<0.0001	
3	0.224	0.225	0.233	0.0058	
4	0.278	0.284	0.292	0.0011	
5	0.395	0.403	0.417	0.0003	

Table 5: Firm characteristics sorted by size and opacity.

6.6 Table 6 and Figure 5: Opacity and R^2

Read:

- Table 6 regresses *IDIOSYN* on *OPAQUE* and controls.
- The coefficient on *OPAQUE* is negative and highly significant, meaning higher opacity is associated with lower idiosyncratic return variation and higher R^2 .
- The positive coefficient on *OPAQUE*² means the relation between opacity and *IDIOSYN* is convex; the effect tails off at high opacity.
- Robustness checks using alternative opacity and return specifications leave the main result intact.
- Figure 5 converts the estimated *IDIOSYN* relation into implied R^2 and shows that R^2 rises with opacity.

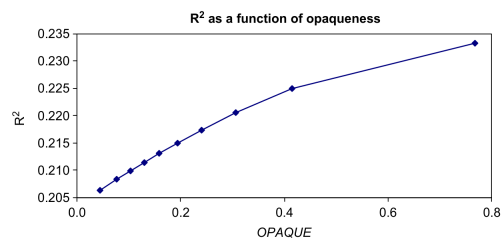
Economic message: Opaque reporting is associated with less firm-specific information entering prices during normal trading periods.

Table 6

Relation between *IDIOSYN* and opacity.

Panel A contains ordinary least square regressions of *IDIOSYN* as a function of opacity and control variables. *IDIOSYN* measures firm-specific information: $IDIOSYN = \ln[(1-R^2)/R^2]$. There are 40,882 firm-years in the sample period 1991–2005. *t*-statistics appear below regression coefficients. Economic impact is the expected change in firm R^2 resulting from an increase in each right-hand-side variable from the 25th to the 75th percentile of the sample distribution. It is derived by calculating the impact on *IDIOSYN* and inverting to estimate the impact on R^2 . Panel B contains robustness checks for coefficients on *OPAQUE* and *OPAQUE*² for variant specifications of Model 3 from Panel A. See Appendix A for all variable definitions.

Panel A Variable	Model 1	Model 2	Model 3	Economic impact (Model 3)
Intercept	2.870 190.22	2.740 177.52	2.782 160.66	
<i>OPAQUE</i>	−0.163 −8.95	−0.167 −9.34	−0.402 −8.49	0.015
Var(Industry Index)	−125.228 −33.49	−124.669 −33.89	−123.064 −33.35	0.019
Size (lagged)	−0.254 −127.44	−0.247 −122.82	−0.249 −121.89	0.157
<i>M</i> to <i>B</i> (lagged)	0.003 2.51	0.003 2.43	0.003 2.67	−0.001
<i>LEV</i> (lagged)	0.001 0.07	0.006 0.35	0.003 0.15	0.000
ROE	−0.022 −2.43	−0.008 −0.94	−0.006 −0.67	0.000
Skewness		0.030 6.17	0.030 6.06	−0.004
Kurtosis		0.051 37.02	0.051 37.14	−0.018
<i>OPAQUE</i> ²			0.228 5.35	−0.003
Adjusted R^2	0.335	0.357	0.357	
Panel B				
Model variation			Coefficient on <i>OPAQUE</i>	Coefficient on <i>OPAQUE</i> ²
1. Baseline regression (Model 3 Panel A)			−0.402 −8.49	0.228 5.35
2. Excluding ROE from set of explanatory variables			−0.401 −8.48	0.230 5.39
3. Requiring 51 weeks of data per year			−0.391 −8.04	0.223 5.02
4. Measuring opacity using single-year discretionary accruals			−0.462 −5.66	0.554 3.44
5. Using biweekly instead of weekly returns in index model			−0.431 −7.73	0.231 4.58
6. Using percentile rank of <i>OPAQUE</i> instead of its numerical value			−0.099 −7.33	
7. By-year (Fama & MacBeth) regressions, full sample			−0.189 −1.72	0.115 1.53
8. By-year (Fama & MacBeth) regressions, excluding post-2002 data			−0.333 −3.31	0.211 3.16



tion of opacity. The horizontal axis value for point on the graph is the average value of *OPAQUE* for decile portfolios formed on the basis of opacity. The vertical axis value is the inferred value of R^2 from regression Model 5 of Table 6, with all other right-hand-side variables set equal to their sample means.

Figure 5: Implied R^2 as a function of opacity.

Table 6: Relation between *IDIOSYN* and opacity.

6.7 Table 7 and Figure 6: Opacity and crash risk

Read:

- Table 7 reports logit regressions predicting firm-specific crashes.
- *OPAQUE* has a positive and highly significant coefficient; *OPAQUE*² has a negative and highly significant coefficient.
- Moving opacity from the 25th to the 75th percentile raises crash probability by about 1.56 percentage points.
- Figure 6 plots annual crash probability by opacity decile. Crash risk rises sharply over the lower and middle opacity range, then flattens.

Economic message: Opaque financial reports predict large negative firm-specific return realizations, consistent with hidden bad news being released suddenly.

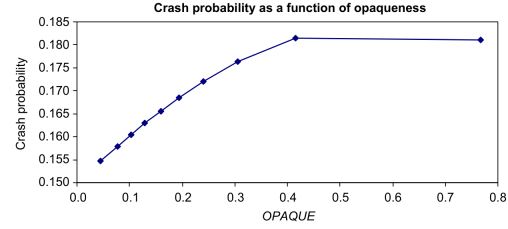
Table 7

Using opacity to predict crash risk.

In Panel A, logit regressions are used to explain crash risk. The dependent variable, *CRASH* is an indicator variable equal to one if within its fiscal year a firm experiences one or more *Firm-Specific Weekly Returns* falling 3.09 or more standard deviations below the mean *Firm-Specific Weekly Return* for its fiscal year and equal to zero otherwise. There are 40,882 firm-years in the sample period 1991–2005. Panel B contains robustness checks for coefficients on *OPAQUE* and *OPAQUE*² for variant specifications of the Panel A regression. See Appendix A for all variable definitions.

Panel A Variable	Coefficient	Chi ²	p-Value	Marginal impact
Intercept	−2.232	1364.49	<0.0001	
<i>OPAQUE</i>	0.855	25.61	<0.0001	
<i>OPAQUE</i> ²	−0.729	22.11	<0.0001	0.0156
<i>ROE</i> _{<i>t</i>}	−0.213	51.73	<0.0001	−0.0047
<i>SIZE</i> _{<i>t</i>−1}	0.118	278.63	<0.0001	0.0477
<i>M</i> to <i>B</i> _{<i>t</i>−1}	0.007	2.85	0.091	0.0020
<i>LEV</i> _{<i>t</i>−1}	−0.399	39.38	<0.0001	−0.0181
Chi ²	369.53	<0.0001		
Number of observations				
Crash	6,950			
No crash	33,932			
Panel B Model variation	Coefficient on <i>OPAQUE</i> (significance level)	Coefficient on <i>OPAQUE</i> ² (significance level)		
1. Baseline regression (as reported in Panel A)	0.855 <0.0001	−0.729 <0.0001		
2. Excluding ROE from set of explanatory variables	0.882 <0.0001	−0.682 <0.0001		
3. Requiring 51 weeks of data per year	0.942 <0.0001	−0.802 <0.0001		
4. Measuring opacity using single-year discretionary accruals	0.800 0.0067	−0.802 <0.0001		
5. Using biweekly instead of weekly returns in index model	0.804 0.0067	−0.773 0.0047		
6. Using percentile rank of <i>OPAQUE</i> instead of its numerical value	0.083 0.088			
7. By-year (Fama & MacBeth) regressions	0.762 <0.0001	−0.702 <0.0001		
8. By-year (Fama & MacBeth) regressions, excluding post-2002 data	0.932 <0.0001	−0.851 <0.0001		

Table 7: Using opacity to predict crash risk.



ash probability as a function of opacity. The horizontal axis value for point on the graph is the average value of *OPAQUE* by ranking on *OPAQUE*. The vertical axis value is the implied probability of a crash during the year for each value of *OPAQUE*, with all other right-hand-side variables set equal to their sample average values.

Figure 6: Annual crash probability as a function of opacity.

6.8 Tables 8 and 9: Sorting evidence and positive-jump test

Read Table 8:

- Crash probability increases with opacity for the three largest size quintiles.
- The high-minus-low opacity difference in crash probability is 3.4, 4.1, and 6.0 percentage points in size quintiles 3, 4, and 5, respectively.
- Raw crash returns are more negative for high-opacity firms.
- Jump probability does not increase with opacity.

Read Table 9:

- Logit regressions show that *OPAQUE* and *OPAQUE*² do not significantly predict positive jumps.
- This contrasts sharply with the crash-risk regressions.

Economic message: Opacity predicts left-tail risk but not right-tail risk. This is the paper's strongest evidence against a generic volatility or fat-tail interpretation.

Table 8

Crash and jump probabilities for firms sorted on size and opacity.

We sort our sample into five size quintiles and into three opacity groups (1 = low; 3 = high). Break points for the two sorts are estimated independently, so the number of observations in each cell differs. *CRASH* is an indicator variable equal to one if within its fiscal year a firm experiences one or more Firm-Specific Weekly Returns falling 3.09 or more standard deviations below the mean Firm-Specific Weekly Return for its fiscal year and equal to zero otherwise. *JUMP* is an indicator variable equal to one if within its fiscal year a firm experiences one or more Firm-Specific Weekly Returns falling 3.09 or more standard deviations above the mean Firm-Specific Weekly Return for its fiscal year and equal to zero otherwise. There are 40,882 firm-years in the sample period 1991–2005.

Panel A: Crash probability for each size/opacity group						
Size	Opacity group			Difference (3)–(1)	p-value (3)–(1)	
	1	2	3			
1	0.117	0.108	0.108	–0.009	0.362	
2	0.151	0.153	0.158	0.007	0.499	
3	0.174	0.184	0.208	0.034	0.001	
4	0.202	0.221	0.243	0.041	<0.0001	
5	0.161	0.181	0.221	0.060	<0.0001	
Panel B: Mean of crash returns (raw one-week stock returns) for each size/opacity group						
Size	Opacity group					Mean
	1	2	3			
1		–0.22	–0.24	–0.26		–0.25
2		–0.23	–0.25	–0.28		–0.26
3		–0.21	–0.23	–0.29		–0.25
4		–0.19	–0.22	–0.26		–0.22
5		–0.15	–0.18	–0.22		–0.17
Mean		–0.19	–0.22	–0.27		
Panel C: Jump probability for each size/opacity group						
Size	Opacity group			Difference (3)–(1)	p-value (3)–(1)	
	1	2	3			
1	0.337	0.358	0.340	0.003	0.813	
2	0.255	0.257	0.247	–0.008	0.521	
3	0.208	0.208	0.198	–0.010	0.348	
4	0.174	0.168	0.161	–0.013	0.227	
5	0.132	0.143	0.123	–0.009	0.367	

Table 8: Crash and jump probabilities sorted by size and opacity.

6.9 Tables 10, 11, and 12: Sarbanes-Oxley interactions

Read Table 10:

Table 9

Using opacity to predict positive jumps.

Logit regressions are used to explain jump probability. The dependent variable, *JUMP*, is an indicator variable equal to one if within its fiscal year a firm experiences one or more Firm-Specific Weekly Returns falling 3.09 or more standard deviations above the mean Firm-Specific Weekly Return for its fiscal year and equal to zero otherwise. There are 40,882 firm-years in the sample period 1991–2005. See Appendix A for all variable definitions.

Variable	Coefficient	Chi ²	p-Value	Marginal impact
Intercept	–0.167	9.39	0.002	
<i>OPAQUE</i>	–0.164	1.15	0.283	
<i>OPAQUE</i> ²	0.172	1.59	0.208	–0.0031
<i>ROE_t</i>	–0.030	1.12	0.290	–0.0007
<i>SIZE_{t–1}</i>	–0.214	962.59	<0.0001	–0.0881
<i>M to B_{t–1}</i>	–0.011	7.06	0.008	–0.0033
<i>LEV_{t–1}</i>	0.221	14.86	0.000	0.0100

Table 9: Using opacity to predict positive jumps.

- The relation between opacity and *IDIOSYN* is strong before SOX.
- The interaction between SOX and opacity has the opposite sign and largely offsets the pre-SOX relation.
- The result suggests that discretionary-accrual opacity became less informative about hidden information after SOX.

Read Table 11:

- Opacity predicts crash risk before SOX.
- The post-SOX interaction terms have the opposite sign, suggesting that the crash-risk relation weakens after SOX.

Read Table 12:

- Opacity does not predict positive jumps either before or after SOX.

Economic message: The SOX evidence is consistent with stronger accounting regulation reducing the ability or incentive to hide bad news through accrual management.

Table 10

Relation between *IDIOSYN* and opacity pre- and post-Sarbanes-Oxley Act. Ordinary least squares regressions of *IDIOSYN* as a function of opacity and control variables. *IDIOSYN* measures firm-specific information. *SOX* is an indicator variable equal to one in 2002 and beyond and equal to zero otherwise. There are 40,882 firm-years in the sample period 1991–2005. *t*-Statistics appear below regression coefficients. See Appendix A for all variable definitions.

Variable	Model 1	Model 2	Model 3
Intercept	2.916 191.14	2.782 179.12	2.833 159.77
<i>SOX</i>	−0.309 −25.06	−0.334 −27.58	−0.399 −23.03
<i>OPAQUE</i>	−0.189 −8.67	−0.189 −8.81	−0.512 −9.20
<i>SOX</i> * <i>OPAQUE</i>	0.233 6.54	0.234 6.70	0.723 7.46
Var(Industry Return)	−135.30 −36.42	−135.76 −37.24	−134.40 −36.76
<i>SIZE</i> (lagged)	−0.243 −120.63	−0.234 −115.78	−0.236 −114.81
<i>M-to-B</i> (lagged)	0.001 1.02	0.001 0.71	0.001 0.89
<i>LEV</i> (lagged)	−0.037 −2.05	−0.035 −1.97	−0.036 −2.04
<i>ROE</i>	−0.030 −3.32	−0.016 −1.77	−0.014 −1.53
Skewness		0.028 5.88	0.028 5.76
Kurtosis		0.054 39.91	0.054 39.99
<i>OPAQUE</i> ²			0.322 6.31
<i>SOX</i> * <i>OPAQUE</i> ²			−0.485 −5.49
Adjusted <i>R</i> ²	0.349	0.374	0.374

Table 10: Relation between *IDIOSYN* and opacity before and after *SOX*.

Table 11

Using opacity to predict crash risk, pre- and post-Sarbanes-Oxley Act (*SOX*).

Logit regressions are used to explain crash risk. The dependent variable, *CRASH*, is an indicator variable equal to one if within its fiscal year a firm experiences one or more *Firm-Specific Weekly Returns* falling 3.09 or more standard deviations below the mean *Firm-Specific Weekly Return* for its fiscal year and equal to zero otherwise. *SOX* is an indicator variable equal to one in 2002 and beyond and equal to zero otherwise. There are 40,882 firm-years in the sample period 1991–2005. See Appendix A for all variable definitions.

Variable	Coefficient	Chi ²	p-Value	Marginal impact
Intercept	−2.285	1284.14	<0.0001	
<i>SOX</i>	0.389	41.61	<0.0001	0.0570
<i>OPAQUE</i>	0.972	21.35	<0.0001	
<i>OPAQUE</i> ²	−0.893	20.14	<0.0001	0.0173
<i>SOX</i> * <i>OPAQUE</i>	−0.622	3.37	0.067	
<i>SOX</i> * <i>OPAQUE</i> ²	0.565	3.23	0.072	−0.0102
<i>ROE</i> _{<i>t</i>}	−0.209	49.08	<0.0001	−0.0046
<i>SIZE</i> _{<i>t</i>−1}	0.105	209.86	<0.0001	0.0421
<i>M to B</i> _{<i>t</i>−1}	0.010	6.11	0.013	0.0030
<i>LEV</i> _{<i>t</i>−1}	−0.357	31.51	<0.0001	−0.0162
Chi ²	477.47	<0.0001		
Number of observations				
Crash	6,950			
No crash	33,932			

Table 11: Opacity and crash risk before and after *SOX*.

Table 12

Using opacity to predict positive jumps, pre- and post-Sarbanes-Oxley Act (SOX).

Logit regressions are used to explain jump probability. The dependent variable, *JUMP*, is an indicator variable equal to one if within its fiscal year a firm experiences one or more *Firm-Specific Weekly Returns* falling 3.09 or more standard deviations above the mean *Firm-Specific Weekly Return* for its fiscal year and equal to zero otherwise. *SOX* is an indicator variable equal to one in 2002 and beyond and equal to zero otherwise. There are 40,882 firm-years in the sample period 1991–2005. See Appendix A for all variable definitions.

	Coefficient	Chi ²	p-Value	Marginal impact
Intercept	−0.169	8.87	0.003	
<i>SOX</i>	0.196	11.20	0.001	0.0570
<i>OPAQUE</i>	−0.266	2.13	0.145	
<i>OPAQUE</i> ²	0.239	2.08	0.149	−0.0056
<i>SOX</i> * <i>OPAQUE</i>	0.108	0.12	0.734	
<i>SOX</i> * <i>OPAQUE</i> ²	−0.116	0.16	0.686	0.0023
<i>ROE</i> _{<i>t</i>}	−0.026	0.87	0.351	−0.0046
<i>SIZE</i> _{<i>t</i>−1}	−0.224	1012.58	<0.0001	0.0421
<i>M</i> to <i>B</i> _{<i>t</i>−1}	−0.009	4.81	0.028	0.0030
<i>LEV</i> _{<i>t</i>−1}	0.248	18.71	<0.0001	−0.0162
Chi ²	1327.45	<0.0001		
Number of observations				
Jump	9,029			
No jump	31,853			

Table 12: Opacity and positive jumps before and after SOX.

7 Short summary

- The paper studies whether opaque financial reporting affects stock-return synchronicity and crash risk.
- Opacity is measured using the prior three-year moving sum of absolute discretionary accruals.
- The sample contains 40,882 U.S. firm-years from 1991 to 2005.
- Opaque firms have higher R^2 and lower *IDIOSYN*, meaning that less firm-specific information is reflected in stock returns.
- Opaque firms are more likely to experience firm-specific stock-price crashes.
- Opacity does not predict positive jumps.
- The asymmetry between crashes and jumps supports a bad-news-hoarding interpretation.
- Restatement evidence validates the opacity proxy.
- The opacity relations weaken after SOX, consistent with stronger reporting regulation reducing earnings-management opacity.
- The paper shows that accounting opacity is related not only to average information flow but also to downside tail risk.

8 Conclusion

8.1 Core conclusion

Hutton, Marcus, and Tehranian show that opaque financial reports are associated with more synchronous stock returns and higher crash risk. Their firm-specific opacity proxy based on discretionary accruals predicts both higher R^2 and greater probability of large negative firm-specific return realizations.

8.2 Economic intuition

The intuition is that managers can temporarily hide bad news through opaque reporting and earnings management. While bad news is hidden, stock prices move less with firm-specific information and more with broad market and industry factors. When accumulated bad news is eventually revealed, the stock can crash.

8.3 Why the positive-jump result matters

The lack of a relation between opacity and positive jumps is crucial. It implies that opacity is not merely a proxy for noisy firms, volatile firms, or firms with generally fat-tailed returns. Instead, opacity is connected to asymmetric downside risk.

8.4 Incremental contribution revisited

The paper gives firm-level evidence for a mechanism that earlier work studied mostly across countries. It also connects accounting research on earnings management to asset-pricing research on R^2 , firm-specific information, and crash risk.

8.5 Implications

The findings imply that investors and risk managers should pay attention to accounting opacity when evaluating tail risk. A firm can appear to have relatively low firm-specific information flow during normal periods while carrying a higher probability of an abrupt negative price adjustment.

8.6 Limitations

The design is observational. Discretionary accruals are an imperfect proxy for intentional opacity, and the SOX test is not a clean natural experiment because other information-environment changes occurred around the same period. The paper's strength is the consistency of the evidence across restatements, R^2 , crashes, positive-jump falsification, and SOX interactions.

8.7 Takeaway

The enduring takeaway is that financial-reporting opacity can hide firm-specific bad news from prices until it is released suddenly.